

Juan Pablo Serrano Pérez

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About

I am a mathematician with interests ranging from pure research and teaching to language learning. Having conducted research in algorithms, I am highly interested in LLMs, Artificial Intelligence, its development, and applications. Currently, I am dedicated to research in pure mathematics, web development, studying Mandarin, and competitive swimming.

Professional Experience

1. Engineer at [Google](#)via Qualitest (Current)

Primarily performing statistical and data analysis for Large Language Models (LLMs). Building auto-raters to assess AI quality and exactness.

2. Software Engineer at ERP Soluciones México (1.5 years)

API services development.

Education

Ph.D. in Mathematics

Center of Research and Advanced Studies (Cinvestav), IPN, Mexico

Advisors: [Dr. Carlos E. Valencia](#) and [Dr. Carlos A. Alfaro](#)

M.Sc. in Mathematical Sciences

Center of Research and Advanced Studies (Cinvestav), IPN, Mexico

Thesis: [Clusters in Optimal Rectilinear Drawings of Complete Graphs: Insights into Potential Recursive Patterns.](#)

Advisor: [Dr. Ruy Fabila Monroy.](#)

B.Sc. in Physics and Mathematics

Escuela Superior de Física y Matemáticas, IPN, Mexico

Research

Research Interests: *My research includes commutative algebra, machine learning, optimization, algorithms, and statistics.*

Academic Work

1. (Book Chapter) [Determinantal Ideals of Graphs](#), 2025

2. [Graphs with two trivial distance ideals over the ring of polynomials with integer coefficients](#), 2025

3. [Distance ideals of digraphs](#), 2025

4. [Evolutive sandpiles](#), 2024.

Projects

My programming projects are mainly built using Python or Sage. They are available [here](#).

Skills

Advanced proficiency in Python 3.x and $\text{\LaTeX}$. Knowledge of MATLAB, Java, C, SageMath, optimization mathematics, and linear/nonlinear optimization. Solid background in probability, statistics, and analysis and design of algorithms:

- Data structures, graph algorithms, randomized algorithms, parallel computing, approximation algorithms, and combinatorial optimization.

Deep Learning Knowledge:

- TensorFlow, Keras, and scikit-learn; convolutional and recurrent neural networks, natural language processing, and Transformers.

Languages

Spanish: Native

English: C1 Proficiency level

Mandarin: Beginner